

TED[15] 5242
(REVISION-2015)

Reg. No.....
Signature.....

DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY/ MANAGEMENT/
COMMERCIAL PRACTICE

SUBJECT NAME: BIOINFORMATICS

[Time : 3 hours]

(Maximum marks : 100)

PART - A

(Maximum marks: 10)

Answer all questions in one or two sentences. Each question carries 2 marks.

1. Define the term bioinformatics?
2. List out the sequence alignment methods.
3. Define DNA microarray?
4. State the need for the identification of protein coding regions?
5. Define molecular structure prediction in bioinformatics? (5x2=10)

PART - B

(Maximum marks: 30)

II Answer any five of the following questions. Each question carries 6 marks.

1. Differentiate between DNA and RNA?
2. Explain central dogma of molecular biology?
3. Explain BLOSUM algorithm?
4. Explain Dot-plot method in sequence alignment?
5. Differentiate between Prokaryotic and Eukaryotic gene structure?
6. Describe the gene expressions in different types of genes?
7. Describe the concept of drug design? (5x6=30)

PART - C

(Maximum marks: 60)

(Answer one full question from each unit. Each full question carries 15 marks)

UNIT – I

III a) With the help of a diagram explain the structure and function of a cell nucleus? (9)

b) Describe various types of RNA with their function? (6)

OR

IV a) Explain the process of protein synthesis in human cell? (9)

b) Write notes on the databases NCBI and Gen Bank? (6)

V a) What is meant by local alignment? How is it different from global alignment? (8)

b) Differentiate between PAM and BLOSUM? (7)

OR

VI a) Explain any one sequence similarity algorithm for local alignment? (7)

b) What is Phylogenetic? Explain phylogenetic tree in detail with an example? (8)

VII a) Explain genomic signal processing? (8)

b) What is DNA spectrogram? Explain the steps involved in generation of DNA spectrogram. (7)

OR

VIII a) Describe the steps involved in microarray fabrication? (8)

b) Explain the process of microarray image analysis? (7)

IX a) Describe about protein treading? (8)

b) Describe about the software BLAST? Write the steps involved in BLAST? (7)

OR

X a) Define the term protein folding? Describe the process of protein folding? (8)

b) Briefly describe the various RNA structures? Discuss about the RNA secondary structure prediction? (7)